

生态研究前沿速览_2026-07-17

生态研究前沿速览

2026-07-17

共收录 37 篇最新且重要研究

【恢复生态】2 篇 | 【海洋与水生】13 篇 | 【环境与气候】9 篇 | 【微生物生态】1 篇 | 【生态网络】1 篇 | 【植物与农业】1 篇 | 【森林与生物多样性】3 篇 | 【保护与生物多样性】6 篇 | 【岛屿与水库生态】1 篇

目录

1. Consequences of agricultural deforestation and subsequent afforestation (农业毁林及后续造林对土壤生物多样性和生态系统多功能性的影响)
2. Mediterranean outflow waters supply North Atlantic convection sites (地中海溢流水供应北大西洋深对流区域)
3. The hydrological asymmetric signature of El Niño-Southern Oscillation (厄尔尼诺-南方涛动对全球陆地水储量的水文非对称信号)
4. Projected future warming induces a long-term loss in global dissolved organic carbon (未来变暖将导致全球海洋溶解有机碳库长期损失)
5. Intercity connectivity enhances urban mobility resilience to extreme rainfall (城市间连通性提升极端降雨下的城市出行韧性)
6. Predictable shifts in microbial species composition lead to community-level environmental stability (微生物物种组成的可预测转变带来群落尺度环境胁迫稳健性)
7. Coastal groundwater phosphorus drives global acceleration of algal blooms (海岸地下水磷输入推动全球藻华加速)
8. Permafrost carbon release scales linearly with overshoot warming media (多年冻土碳释放随超调变暖线性增加并受 AMOC 临界变化调节)
9. Structural and spectral adaptation of the seagrass *Posidonia oceanica* (海草 *Posidonia oceanica* 光系统 I 对海底光环境的结构与)
10. Core reorganization in food webs buffers ecosystems against warming (食物网核心重组缓冲生态系统对变暖的响应)
11. Food systems transformation would reshape global agriculture (食物系统转型将重塑全球农业格局)
12. The economically optimal mix and timing of coastal adaptation in Europe (欧洲至 2150 年的海岸适应最优组合与时机)
13. Future flood risk concentration for residents near small rivers across the US (美国小河流沿岸居民未来洪水风险更集中)
14. Preferential nitrogen retention characterizes current eutrophication in the US (美国湖泊当前富营养化表现为氮优先滞留)

15. Scaling biodiversity-stability relationships from populations to meta- (跨营养级尺度上从种群到集合群落的生物多样性-稳定性关系)
16. Increasing forest disturbance enhances habitat suitability for Europe' (森林扰动增加欧洲大型食草动物栖息地适宜性)
17. A unified analysis of global riverine eDNA reveals common associations (全球河流 eDNA 统一分析揭示鱼类多样性与流域特征的共同关联)
18. Applying ecological thresholds to inform conservation and restoration (利用生态阈值指导溪流鱼类保护与修复)
19. Stricter protected areas in China coincide with more firm openings and (中国严格保护地与企业进退动态及更高自然效益同时出现)
20. Global high-resolution mapping of seagrass to support conservation (全球高分辨率海草地图为保护提供新基线)
21. Long-term multiple global change interactions amplify belowground carb (长期多重全球变化交互作用放大地下碳分配)
22. Ecological integrity of avoided deforestation projects (避免毁林项目的生态完整性成效并不稳定)
23. Temperate local extinctions from climate change are outpacing tropical (气候变化导致的温带局地灭绝速度超过热带)
24. Globally constrained forest biophysical cooling benefits under rising (大气干燥度上升在全球范围限制森林生物物理降温效益)
25. A fixed methane filter maximizes freshwater emissions under warming (固定效率的甲烷过滤器使淡水升温下排放最大化)
26. Ninety-year trends reveal sharpest insect declines in the mid-twentieth (九十年趋势揭示昆虫最剧烈下降发生于 20 世纪中叶)
27. Amplified Arctic iceberg traffic reshapes benthic biodiversity (增强的北极冰山输运重塑底栖生物多样性)
28. Mining triggers extensive additional deforestation in sub-Saharan Africa (采矿在撒哈拉以南非洲触发大范围额外毁林)
29. A 5.3-million-year-old deep-sea whale necropolis in the Diamantina Zone (迪亚曼蒂纳带发现延续 530 万年的深海鲸落墓地)
30. Ecological multifunctionality of watersheds increases with tree species (流域生态多功能性随树种丰富度增加)
31. Universal dose-response sensitivity of ecological processes to low level (生态过程对低水平夜间人工光呈普遍剂量响应敏感性)
32. Multi-species modelling to improve conservation decision-making for ungulates (多物种模型改善研究不足类群的保护决策)
33. Beyond species richness: Habitat fragmentation reduces community occupancy (超越物种丰富度: 破碎化降低山地森林哺乳动物群落占域和功能丰富度)
34. Several small reservoir islands consistently support higher taxonomic diversity (多个小型水库岛屿在多类群和多维度上持续承载更高多样性)
35. Compounding impacts of riparian hardening and navigation erode phytoplankton (岸线硬化与航运的叠加影响削弱大型河流浮游植物群落韧性)
36. Conservation and natural recovery limits of Madagascar's ultramafic rainforest (马达加斯加超镁铁质雨林的保护价值与自然恢复上限)
37. Linking trophic guild function to biodiversity and ecosystem resilience (营养功能群连接河流—洪泛区—海岸连续体的多样性与韧性)

恢复生态

1. Consequences of agricultural deforestation and subsequent afforestation on soil biodiversity and ecosystem multifunctionality

(农业毁林及后续造林对土壤生物多样性和生态系统多功能性的影响)

Nature Communications | 2026-07-17 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

A 405 paired-plot transect across tropical, subtropical, temperate and boreal zones shows that agricultural deforestation reduces soil multitrophic biodiversity by about 25% and ecosystem multifunctionality by about 58%. Subsequent afforestation partially restores soil biota and functions, but does not return systems to pristine levels, and recovery of biodiversity-function relationships is stronger in warmer zones.

中文摘要

研究沿热带、亚热带、温带和寒带的南北样带调查 405 组成对样地，发现农业毁林使土壤多营养级生物多样性下降约 25%，生态系统多功能性下降约 58%。后续造林可部分恢复土壤生物和功能，但尚未达到原始水平；生物多样性-多功能性关系的恢复在暖区更明显。

深度解读

这项研究把造林恢复的成效从植被覆盖推进到土壤食物网、微生物组和生态系统功能层面。它提醒我们，造林并不自动等于生态恢复，原始土壤生物网络和功能耦合可能存在长期滞后。暖区更强的恢复信号说明气候背景会调节恢复路径，不能用单一造林指标评估不同生物群区。对退化地修复而言，土壤多营养级监测应成为成效评价的核心指标。

DOI: <https://doi.org/10.1038/s41467-026-75740-z>

海洋与水生

2. Mediterranean outflow waters supply North Atlantic convection sites

(地中海溢流水供应北大西洋深对流区域)

Nature Communications | 2026-07-17 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

More than two decades of Argo salinity and oxygen profiles show that Mediterranean Outflow Water can be traced into subpolar North Atlantic basins, including the Irminger Sea. The warm, saline outflow may precondition deep convection sites that contribute to water-mass formation for the lower limb of the Atlantic Meridional Overturning Circulation.

中文摘要

研究利用二十余年的 Argo 盐度和氧气剖面数据，追踪地中海溢流水向亚极地北大西洋的北向输送，确认其可进入包括伊明格海在内的深对流海盆。该暖盐水团可能预调控深对流区，从而影响参与大西洋经向翻转环流下支的水团形成。

深度解读

这篇文章的重要性在于为北大西洋深对流的水团来源提供了更清晰的观测证据。地中海溢流水若能进入亚极地对流区，就会把区域性盐度异常与大尺度环流稳定性联系起来。对海洋生态而言，翻转环流变化会通过温度、营养盐、氧和碳输送改变生产力格局。该结果也说明长期 Argo 观测对识别慢变量水团通道不可替代。

DOI: <https://doi.org/10.1038/s41467-026-75523-6>

环境与气候

3. The hydrological asymmetric signature of El Niño-Southern Oscillation on global water storage

(厄尔尼诺-南方涛动对全球陆地水储量的水文非对称信号)

Communications Earth & Environment | 2026-07-17 | **【Latest】**



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

A machine-learning reconstruction of terrestrial water storage anomalies from 1984 to 2023 identifies 81 El Niño-related and 85 La Niña-related clusters across more than 105 million km². La Niña affects about 20% more land area than El Niño, drying occurs across nearly half of ENSO-affected regions, and wet and dry responses are not locally compensatory.

中文摘要

研究用机器学习重建 1984-2023 年陆地水储量异常，识别出 81 个厄尔尼诺相关和 85 个拉尼娜相关区域，覆盖超过 1.05 亿平方千米。拉尼娜影响陆地面积比厄尔尼诺高约 20%，近半受 ENSO 影响区域出现干化，湿化和干化响应在空间上并不相互抵消。

深度解读

这项研究把 ENSO 影响从降水异常推进到陆地水储量这一生态水文关键变量。非对称性意味着用平均 ENSO 响应评估旱涝风险会低估地方生态压力。干区快速响应并放大亏缺，湿区则具有更强水文记忆，这会影响植被恢复、河流基流和水库调度。它也为把季节气候预测转化为流域生态风险预警提供了新指标。

DOI: <https://doi.org/10.1038/s43247-026-03806-3>

海洋与水生

4. Projected future warming induces a long-term loss in global dissolved organic carbon pool

(未来变暖将导致全球海洋溶解有机碳库长期损失)

Communications Earth & Environment | 2026-07-17 | **【Latest】**



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

An Earth system model with four dissolved organic carbon pools of varying reactivity projects substantial future decreases across the global ocean dissolved organic carbon pool under warming. Losses occur under both weak and strong warming and are irreversible over centennial timescales, with potential consequences for remineralized nutrients and primary production.

中文摘要

研究在地球系统模型中加入四类反应性和寿命不同的溶解有机碳库，预测未来变暖将导致全球海洋溶解有机碳显著减少。无论弱变暖还是强变暖情景，模型均显示该碳库会发生百年尺度上难以逆转的长期损失，并可能改变再矿化营养盐和初级生产。

深度解读

海洋溶解有机碳是巨大而慢周转的碳库，其变化常被低估。该文提示变暖不仅影响表层生产和碳泵强度，也可能改变长期有机碳储存结构。若难降解碳库下降并转化为再矿化营养盐，未来生产力和缺氧格局都可能发生反馈。对全球碳循环模型而言，溶解有机碳的分库表示和持续观测是关键短板。

DOI: <https://doi.org/10.1038/s43247-026-03809-0>

环境与气候

5. Intercity connectivity enhances urban mobility resilience to extreme rainfall

(城市间连通性提升极端降雨下的城市出行韧性)

Nature Climate Change | 2026-07-17 | 【Latest】

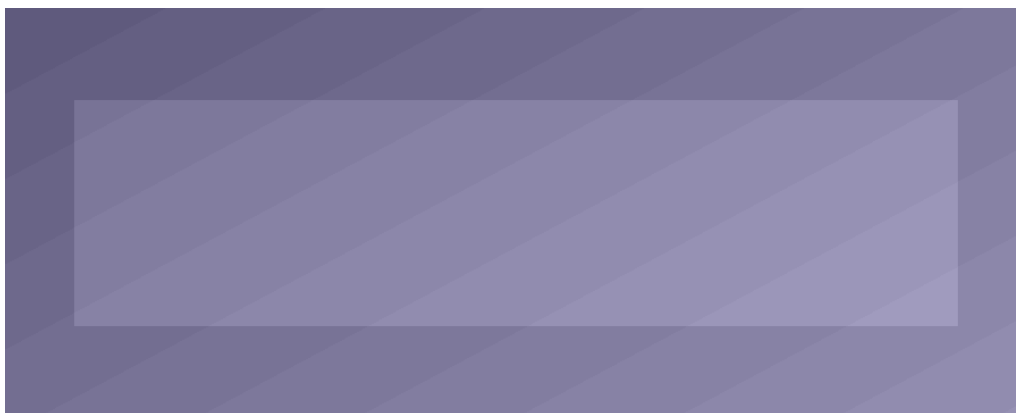


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Using mobility data from 1.1 billion devices across 366 Chinese cities, the study shows that stronger intercity connectivity improves intracity mobility resilience during extreme rainfall. Cities with higher network centrality and links to wealthier cities recover faster, rebound more strongly and suffer lower total performance loss; future rail and socioeconomic development could reduce median city-level losses by about 72.5-76.1%.

中文摘要

研究基于中国 366 个城市、11 亿台设备的移动数据，发现城市间连通性越强，极端降雨期间城市内部出行韧性越高。网络中心性更高、与较富裕城市联系更强的城市恢复更快、反弹更强、总功能损失更低；到 2050 年，铁路和社会经济发展可能使城市层面中位损失下降约 72.5-76.1%。

深度解读

这篇文章把气候适应从单城防灾能力扩展到城市网络韧性。极端降雨通常造成局地冲击，但恢复资源、信息和人员流动具有跨城市属性。对区域生态环境治理而言，交通连通性既能提高救灾效率，也可能改变暴露和扩张压力，因此需要与海绵城市、河湖缓冲空间和土地管控共同设计。小城市受益更大，说明网络投资也具有气候公平含义。

DOI: <https://doi.org/10.1038/s41558-026-02713-x>

微生物生态

6. Predictable shifts in microbial species composition lead to community-wide robustness to environmental stress

(微生物物种组成的可预测转变带来群落尺度环境胁迫稳健性)

Nature Microbiology | 2026-07-17 | 【Latest】

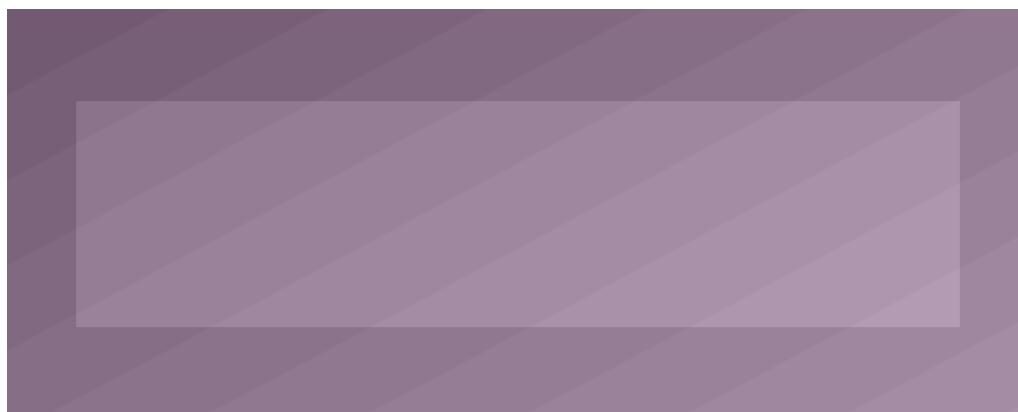


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Experiments with natural aquatic microbial communities across salinity gradients show that increasing salinity shifts communities toward faster-growing species. Although most individual isolates grow more slowly under salinity stress, community mean growth remains more robust than expected; pairwise competitions, 16S data and generalized Lotka-Volterra modelling support this mechanism.

中文摘要

研究在不同盐度下培养天然水生微生物群落，发现盐度胁迫会使群落组成转向生长更快的物种。尽管多数单个菌株在高盐下生长变慢，群落平均生长率却比单物种预测更稳健；双物种竞争、16S 数据和广义 Lotka-Volterra 模型共同支持这一机制。

深度解读

这项工作解释了为什么微生物群落在环境恶化时仍可维持部分功能。它把群落稳健性归因于组成重排，而不是单个物种耐受性的简单平均。对盐水入侵、河口变化和土壤盐渍化研究而言，这意味着功能维持可能伴随隐蔽的多样性损失。管理和模型预测需要同时追踪群落功能与物种替换，否则会误判系统健康。

DOI: <https://doi.org/10.1038/s41564-026-02422-3>

海洋与水生

7. Coastal groundwater phosphorus drives global acceleration of algal blooms

(海岸地下水磷输入推动全球藻华加速)

Nature Communications | 2026-07-16 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

A global physical-chemical analysis links recent coastal algal bloom acceleration more strongly to dissolved inorganic phosphorus supplied by coastal groundwater than to nitrogen availability or water-column stability. Anoxic groundwater releases phosphorus and can explain persistent blooms in regions where nitrogen inputs have been reduced.

中文摘要

研究结合过程框架和全球数据，发现近几十年海岸藻华加速与海岸地下水输入的溶解无机磷关系更强，而不是主要由氮供给或水体稳定性解释。缺氧地下水可释放大量磷，解释一些地区即使控氮后藻华仍持续或加剧的现象。

深度解读

该文直接挑战了许多海岸富营养化治理中以控氮为中心的默认思路。地下水作为非点源、慢响应和难监测路径，可能长期维持近岸磷负荷。治理上需要把地下水氧化还原状态、岸带开发、污水入渗和氮磷协同控制纳入同一框架。对藻华预警而言，仅看河流和表层水营养盐会漏掉关键驱动。

DOI: <https://doi.org/10.1038/s41467-026-75420-y>

环境与气候

8. Permafrost carbon release scales linearly with overshoot warming mediated by AMOC tipping

(多年冻土碳释放随超调变暖线性增加并受 AMOC 临界变化调节)

Nature Communications | 2026-07-16 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Three climate models of varying complexity indicate that permafrost carbon loss during temperature overshoot scales linearly with cumulative warming exposure, at roughly 11-21 PgC per 100 degree-years. A temporary AMOC slowdown partly offsets permafrost warming through Northern Hemisphere cooling, but does not remove broader adverse climate and societal impacts.

中文摘要

研究使用三个复杂度不同的气候模型评估温度超调情景下多年冻土碳响应，发现冻土碳损失与累积变暖暴露近似线性相关，约为每 100 度年损失 11-21 PgC。暂时性 AMOC 减弱可通过北半球相对降温部分抵消冻土变暖，但并不能消除气候和社会负面影响。

深度解读

这项研究把超调变暖的风险转化为可用于碳预算讨论的累积暴露指标。即便后期降温，冻土碳损失仍可能留下长期痕迹，说明净零路径不能只看 2100 年温度目标。AMOC 减弱带来的局部冷却不是安全缓冲，因为它本身会触发其他风险。对气候政策而言，减少超调幅度和持续时间比事后依赖大规模碳移除更稳妥。

DOI: <https://doi.org/10.1038/s41467-026-73612-0>

海洋与水生

9. Structural and spectral adaptation of the seagrass *Posidonia oceanica* photosystem I to seabed light

(海草 *Posidonia oceanica* 光系统 I 对海底光环境的结构与光谱适应)

Nature Communications | 2026-07-16 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Cryo-electron microscopy and ultrafast spectroscopy reveal how the seagrass *Posidonia oceanica* adapts photosystem I to dim, spectrally shifted seabed light. Expanded antenna structures and loss of low-energy chlorophyll forms reduce exciton trapping time, enhancing photon-use efficiency, and reversion to land-plant ortholog sequences restores red-shifted emission.

中文摘要

研究通过冷冻电镜和超快光谱揭示海草 *Posidonia oceanica* 如何适应昏暗且光谱偏移的海底光环境。其光系统 I 扩大天线结构，同时丢失低能叶绿素形式以缩短激子捕获时间，提高弱光下光子利用效率；恢复陆生植物同源序列可重新产生红移发射。

深度解读

海草草甸是重要蓝碳生态系统，但其弱光适应机制长期缺乏分子解释。该文把海草从陆地重返海洋后的演化适应具体落实到光系统结构。对蓝碳保护而言，理解光利用效率有助于解释浑浊、水深和附生生物压力下的草甸退化。它还为改良作物弱光利用提供了跨生态系统启发。

DOI: <https://doi.org/10.1038/s41467-026-75200-8>

生态网络

10. Core reorganisation in food webs buffers ecosystems against warming

(食物网核心重组缓冲生态系统对变暖的响应)

Communications Earth & Environment | 2026-07-16 | **【Latest】**



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Across 14 whole-stream ecosystems along an Icelandic natural temperature gradient, warming did not change the number or proportion of core and peripheral species, but it reorganized the identity and abundance of core species. Although warming reduced robustness through lower connectance, core reorganization enhanced energy flux and robustness to species loss.

中文摘要

研究利用冰岛 14 个完整溪流生态系统的自然温度梯度，发现变暖并未改变核心和边缘物种的数量或比例，却重组了核心物种的身份和丰度。尽管变暖因连接度下降而削弱次级灭绝稳健性，核心重组仍提高了能流和物种损失稳健性。

深度解读

这篇文章为食物网如何在变暖中维持功能提供了机制证据。它说明生态系统韧性不只来自物种数，也来自关键网络位置上的物种替换和丰度调整。若核心类群丧失，模型显示生态系统会不成比例地受损，因此监测应关注网络核心而非平均多样性。对溪流和高纬生态系统管理而言，保护核心能流通道可能比单纯保护物种清单更关键。

DOI: <https://doi.org/10.1038/s43247-026-03813-4>

植物与农业

11. Food systems transformation would reshape global agriculture

(食物系统转型将重塑全球农业格局)

Nature | 2026-07-15 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

A multimodel ensemble of ten global economic models evaluates a food-system transformation scenario combining healthy diets, higher productivity and a halving of food loss and waste by 2050. The scenario implies a 6% median decrease in agricultural land compared with 2020, 17% lower agricultural production than business-as-usual projections, and a major shift away from livestock production value toward vegetables, fruits, nuts and legumes.

中文摘要

研究使用十个全球经济模型组成的多模型集合，评估到 2050 年健康膳食、生产率提升和食物损失浪费减半共同发生时的全球农业变化。结果显示，农业用地相对 2020 年中位数下降 6%，农业产量相对照常情景下降 17%，畜牧业产值显著低于原有预测，而蔬菜、水果、坚果和豆类产值上升。

深度解读

这篇文章把食物系统转型从环境倡议转化为可量化的农业结构冲击。它说明，健康和可持续饮食并不只是消费端选择，而会沿供应链改变土地利用、生产部门和区域经济。管理上需要提前处理畜牧业转型、农户收入、营养安全和生态减压之间的分配问题。对生态研究而言，关键价值在于将土地需求下降与生物多样性恢复、碳汇和水资源压力缓解联系起来。

DOI: <https://doi.org/10.1038/s41586-026-10775-2>

环境与气候

12. The economically optimal mix and timing of coastal adaptation in Europe to 2150

(欧洲至 2150 年的海岸适应最优组合与时机)

Nature Communications | 2026-07-15 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

The study develops economically optimal adaptation pathways for 41,327 European coastal floodplains through 2150, considering protection, retreat, accommodation and combinations of these options under multiple climate scenarios. For 95% of the coastline requiring adaptation, initial action is economically optimal immediately, while retreat is optimal for 22% of coastline by 2150 and protection for 9%. Adaptation tipping points arrive earlier under higher climate-change scenarios.

中文摘要

研究为欧洲 41,327 个海岸洪泛区构建到 2150 年的经济最优适应路径，比较防护、后退、适应性建筑改造及其组合。需要适应的海岸线中，95% 的最优初始行动应立即启动；到 2150 年，22% 的海岸线经济上更适合后退，9% 更适合防护。高排放情景会使适应转折点平均提前。

深度解读

海岸适应常被简化为修堤或撤退的二选一，而这篇研究强调组合、顺序和时机同样决定成本。其管理含义是，拖延会锁定资产和人口暴露，使未来后退更昂贵。经济最优并不等同于社会可接受或生态最优，因此实际规划还需叠加湿地保护、公平补偿和地方治理能力。该框架适合用于筛选优先区，再进入地方尺度的多目标评估。

DOI: <https://doi.org/10.1038/s41467-026-74042-8>

环境与气候

13. Future flood risk concentration for residents near small rivers across the USA

(美国小河流沿岸居民未来洪水风险更集中)

Nature Sustainability | 2026-07-10 | **【Recent】**



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

By integrating high-resolution hydrodynamic simulations, demographic projections and current flood-defence levels across the contiguous United States, the study shows that small rivers are especially sensitive to climate-change-driven increases in flood intensity by 2050. Existing defences along small rivers are comparatively weak, and restoring future risk to historical levels could require the highest defence investments among river sizes, reaching about US\$4 million per kilometre in some locations.

中文摘要

研究整合美国本土高分辨率水动力模拟、人口预测和现有防洪标准，发现到 2050 年，小河流对气候变化导致的洪水强度增加更敏感，而其沿岸防洪水平又普遍较弱。若要把未来风险降回历史水平，小河流沿线可能需要最高的单位长度防洪投资，部分地区约达每千米 400 万美元。

深度解读

该文的重要性在于把洪水风险从大江大河转向更容易被忽视的小河流和社区尺度。小河流风险上升主要由气候危害增强和防护不足驱动，而大河流风险更受暴露增长控制，这意味着两类河流不能套用同一投资逻辑。对流域生态管理而言，小河流防洪若只依赖硬工程，可能进一步压缩河岸带和洪泛区功能。更合理的路径应把防洪投资、自然岸线恢复、洪泛区退让和脆弱社区保护放在一起评估。

DOI: <https://doi.org/10.1038/s41893-026-01896-7>

海洋与水生

14. Preferential nitrogen retention characterizes current eutrophication in United States lakes

(美国湖泊当前富营养化表现为氮优先滞留)

Nature Communications | 2026-07-09 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

A long-term analysis of 121 United States lakes from 2012 to 2022 shows that nutrient retention has shifted from phosphorus-dominated to preferential nitrogen retention. The authors identify nutrient dynamics resembling critical flickering, suggesting that some lakes may be approaching a different eutrophication regime.

中文摘要

研究追踪美国 121 个湖泊 2012-2022 年的氮、磷滞留变化，发现湖泊营养盐循环从磷优先滞留转向氮优先滞留，并出现类似临界闪烁的动态信号。这表明部分湖泊可能不是简单的富营养化复发，而是在接近或进入新的营养循环状态。

深度解读

这篇文章的重要性在于挑战了长期湖泊治理中过度依赖控磷的惯性。氮优先滞留意味着沉积物、反硝化、藻类群落和水动力过程可能正在共同改变系统反馈。对恢复管理而言，氮磷协同监测、内源释放评估和状态转换预警应成为常规工具。它也提示，治理成效不能只看短期水质改善，还要判断湖泊是否已形成新的稳定态。

DOI: <https://doi.org/10.1038/s41467-026-75318-9>

森林与生物多样性

15. Scaling biodiversity-stability relationships from populations to meta-communities across trophic levels
(跨营养级尺度上从种群到集合群落的生物多样性-稳定性关系)

Nature Communications | 2026-07-07 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Using a large forest biodiversity experiment, the study tests how biodiversity-stability relationships scale from populations to local communities and meta-communities in a plant-herbivore system. Higher diversity generally stabilizes ecological dynamics, and herbivore diversity can feed back to stabilize plant populations and communities.

中文摘要

研究基于大型森林生物多样性实验，检验植物-植食性昆虫系统中稳定性如何从种群、局地群落扩展到集合群落。结果显示，较高多样性总体上具有稳定作用，植食者多样性还可通过自上而下反馈稳定植物种群和群落。

深度解读

这项研究把生物多样性-稳定性关系从单一营养级推进到真实食物网。它提醒森林恢复不能只看树种数，也需要追踪植食昆虫和天敌等中间环节。若昆虫多样性下降，植物群落稳定性可能被间接削弱。实验尺度上的证据也有助于改进多营养级生态系统模型。

DOI: <https://doi.org/10.1038/s41467-026-75366-1>

森林与生物多样性

16. Increasing forest disturbance enhances habitat suitability for Europe's large herbivores

(森林扰动增加欧洲大型食草动物栖息地适宜性)

Nature Ecology & Evolution | 2026-06-26 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Tracking data from 3,069 individuals of European bison, moose, red deer and roe deer are combined with satellite disturbance maps. The analysis finds that forest disturbance can increase habitat suitability for large herbivores for up to 35 years, with responses shaped by patch size and alternative foraging habitats.

中文摘要

研究整合欧洲野牛、驼鹿、赤鹿和狍等 3069 个个体的追踪数据与卫星扰动图，发现森林扰动后长达 35 年内，大型食草动物倾向选择扰动斑块。斑块大小和周边替代觅食地会调节这种选择强度。

深度解读

该文避免了把森林扰动简单等同于栖息地损失的线性判断。对大型食草动物而言，扰动可创造光照充足、草本和灌木丰富的觅食空间。管理上需要区分自然扰动、商业采伐和灾后清理，因为三者对结构复杂性和人兽冲突的影响不同。它也为欧洲再野化和森林多目标经营提供了更细的证据。

DOI: <https://doi.org/10.1038/s41559-026-03096-0>

海洋与水生

17. A unified analysis of global riverine eDNA reveals common associations of fish biodiversity with drainage characteristics

(全球河流 eDNA 统一分析揭示鱼类多样性与流域特征的共同关联)

Nature Ecology & Evolution | 2026-06-24 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

A harmonized analysis of environmental DNA data from 113 river systems examines global fish biodiversity patterns and their associations with drainage characteristics and human pressures. The study demonstrates the value of eDNA for comparable large-scale river biodiversity monitoring.

中文摘要

研究利用 113 个河流系统的环境 DNA 数据，统一分析全球河流鱼类多样性格局及其与流域属性、人类活动压力的关系。结果展示了 eDNA 在跨区域鱼类多样性监测中的可比性和尺度优势。

深度解读

eDNA 正在从补充调查工具转向河流生物多样性监测的核心技术。统一分析的价值在于减少不同研究之间采样、引物和数据库差异带来的不可比性。应用时仍需重视阴性对照、参考库缺口和传统调查校准。尤其不能把 DNA 检出直接等同于稳定种群存在。

DOI: <https://doi.org/10.1038/s41559-026-03106-1>

海洋与水生

18. Applying ecological thresholds to inform conservation and restoration efforts for stream fishes

(利用生态阈值指导溪流鱼类保护与修复)

npj Biodiversity | 2026-06-27 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

The study applies land-use pressure thresholds to approximately 1.73 million stream segments in the United States and Europe. By combining threshold status with protected-area coverage, it distinguishes preventive conservation priorities from restoration priorities for stream fishes.

中文摘要

研究将土地利用压力阈值应用于美国和欧洲约 173 万个河流段，并结合保护地覆盖率，识别接近阈值的预防保护区和已越阈的修复优先区。这为溪流鱼类保护提供了空间化决策框架。

深度解读

该框架把“应该保护还是修复”转化为可以落图的管理问题。阈值方法的优势是直观，但阈值本身应附带不确定区间，而不是被当成绝对红线。溪流鱼类响应还受河网连通性、水文改变、污染和生活史差异影响。实际规划需要把阈值筛选作为第一步，再用本地调查和压力源诊断细化。

保护与生物多样性

19. Stricter protected areas in China coincide with more firm openings and closures and higher nature benefits
(中国严格保护地与企业进退动态及更高自然效益同时出现)

Communications Sustainability | 2026-06-30 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

An analysis of more than 50 million business records and 6,638 protected areas in China links stricter protected-area policies with changes in firm openings and closures and with higher ecosystem-service values. The effects vary across regions and policy contexts.

中文摘要

研究分析 5000 余万条企业记录和 6638 个中国保护地，发现更严格保护政策与企业进入、退出动态变化及较高生态系统服务价值相关，但不同地区存在权衡。结果为保护地政策的社会经济与生态协同评估提供了大样本证据。

深度解读

这篇文章的亮点是把保护地生态效益与企业微观动态放在同一框架下分析。它提示保护政策不能只看企业数量变化，更要看产业结构调整 and 自然收益是否同步出现。由于研究主要是观察性关联，不能把所有变化直接解释为严格管制的因果效应。后续需要准实验设计识别政策冲击。

DOI: <https://doi.org/10.1038/s44458-026-00108-9>

海洋与水生

20. Global high-resolution mapping of seagrass to support conservation
(全球高分辨率海草地图为保护提供新基线)

Nature | 2026-06-24 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41586-026-10704-3>

Abstract

Using 4.75 million Sentinel-2 images and deep learning, the study produces the first global 10-m maps of clear-water seagrass extent for 2019–2020 and 2023–2024. It maps 148,506 km² and documents widespread recent loss and degradation, much of it outside protected areas.

中文摘要

研究利用 475 万幅 Sentinel-2 影像和深度学习，首次构建全球 10 米分辨率的清水海域海草分布及变化图。结果识别出 148,506 平方千米海草，并显示 2019 年以来存在广泛损失和退化，许多海草床位于保护地之外。

深度解读

这是海草保护从零散样点走向全球一致监测的重要基础设施。高分辨率变化图可直接支持保护空缺识别、蓝碳核算和修复优先级排序。需要注意，浑浊水体和深水海草仍可能被低估，后续应结合声学在现场调查校准。

DOI: <https://doi.org/10.1038/s41586-026-10704-3>

环境与气候

21. Long-term multiple global change interactions amplify belowground carbon allocation

(长期多重全球变化交互作用放大地下碳分配)

Nature Climate Change | 2026-06-23 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41558-026-02678-x>

Abstract

An 11-year multifactor grassland experiment tested warming, rainfall reduction, elevated CO₂ and nitrogen addition. Warming and elevated CO₂ increased total belowground carbon allocation, while drought and nitrogen altered the size and persistence of these responses.

中文摘要

一项持续 11 年的草地多因子实验同时检验增温、减雨、CO₂ 升高和氮添加。增温与 CO₂ 升高增加地下总碳分配，但干旱和氮沉降会改变这一响应的幅度与持续性。

深度解读

长期实验的价值在于揭示单因子短期试验难以看到的交互效应。地下碳输入增加并不自动等于稳定土壤碳库，因为根系周转和微生物分解也会同步变化。陆地碳模型应显式表示多因子共同作用及其随时间变化。

DOI: <https://doi.org/10.1038/s41558-026-02678-x>

保护与生物多样性

22. Ecological integrity of avoided deforestation projects

(避免毁林项目的生态完整性成效并不稳定)

Nature Climate Change | 2026-06-22 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41558-026-02657-2>

Abstract

The authors compare 133 avoided-deforestation projects with matched controls to test whether carbon projects also safeguard forest ecological integrity. Outcomes are highly variable, and many projects show negligible, mixed or negative ecological effects.

中文摘要

研究将 133 个避免毁林项目与匹配对照区比较，评估碳项目是否也保护森林生态完整性。结果高度不一致，许多项目的生态效应不显著、正负混合，甚至为负。

深度解读

碳信用不能被直接当作生物多样性收益的代理指标。项目核证需要同时监测森林结构、连通性和物种组成，而不仅是树冠或碳量。该结果也提示自然气候解决方案必须设置独立的生态完整性保障条款。

DOI: <https://doi.org/10.1038/s41558-026-02657-2>

环境与气候

23. Temperate local extinctions from climate change are outpacing tropical extinctions
(气候变化导致的温带局地灭绝速度超过热带)

Nature Climate Change | 2026-06-18 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41558-026-02669-y>

Abstract

A global resurvey synthesis covering 5,151 species finds climate-linked local extinctions more frequent in temperate than tropical regions. Faster warming at higher latitudes, rather than uniformly greater biological sensitivity, helps explain the pattern.

中文摘要

全球尺度的重复调查综合分析覆盖 5,151 个物种，发现与气候变化相关的局地灭绝在温带比热带更常见。较高纬度更快的增温，而非一致更高的生物敏感性，是解释该格局的重要因素。

深度解读

这一结果修正了‘热带物种总是最脆弱’的简单叙事。风险取决于暴露速度、极端事件和物种耐受性共同作用。保护规划应同时守住热带高敏感物种与温带快速变化前沿。

DOI: <https://doi.org/10.1038/s41558-026-02669-y>

环境与气候

24. Globally constrained forest biophysical cooling benefits under rising atmospheric dryness
(大气干燥度上升在全球范围限制森林生物物理降温效益)

Nature Climate Change | 2026-06-17 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41558-026-02677-y>

Abstract

Global observations from 2001–2023 show that vapour-pressure deficit is a primary constraint on forest biophysical cooling relative to nearby open land. Rising atmospheric dryness can weaken the non-carbon climate benefits of forests.

中文摘要

2001—2023 年的全球观测表明，饱和水汽压亏缺是限制森林相对邻近开阔地降温效应的关键因素。大气持续变干可能削弱森林除固碳之外的气候调节收益。

深度解读

森林气候效益不能只用碳汇衡量，蒸散与反照率同样关键。大气干燥会促使气孔关闭，使森林降温能力下降。造林和森林管理的气候评估需要纳入区域水分约束与树种水力策略。

DOI: <https://doi.org/10.1038/s41558-026-02677-y>

海洋与水生

25. A fixed methane filter maximizes freshwater emissions under warming

(固定效率的甲烷过滤器使淡水升温下排放最大化)

Nature Climate Change | 2026-06-05 | **【Recent】**



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41558-026-02649-2>

Abstract

Freshwater warming experiments show that methane production rises while the microbial methane-oxidation filter does not strengthen proportionally. The resulting imbalance increases methane emissions from warming freshwaters.

中文摘要

淡水增温实验显示，甲烷产生随温度升高而增强，但微生物甲烷氧化过滤并未同比加强。二者失衡导致变暖淡水系统向大气释放更多甲烷。

深度解读

淡水是气候反馈中面积小但通量高的关键环节。若氧化过滤效率缺乏可塑性，升温会把更多新生成甲烷直接转化为排放。湖泊和河流碳预算应同时测量产生与氧化过程，而不能只测表面通量。

DOI: <https://doi.org/10.1038/s41558-026-02649-2>

保护与生物多样性

26. Ninety-year trends reveal sharpest insect declines in the mid-twentieth century

(九十年趋势揭示昆虫最剧烈下降发生于 20 世纪中叶)

Nature Ecology & Evolution | 2026-06-02 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41559-026-03074-6>

Abstract

Using 1.2 million Swiss records of 595 saproxylic beetle species, the study reconstructs insect richness over nine decades. The steepest declines occurred during mid-twentieth-century agricultural intensification, revealing a strong shifted-baseline problem.

中文摘要

研究基于瑞士 595 种腐木甲虫的 120 万条记录重建九十年物种丰富度变化。最陡峭的下降发生在 20 世纪中叶农业集约化时期，显示现代监测存在明显的基线漂移。

深度解读

只看近几十年数据可能低估历史损失，也会把已经贫化的群落误当作正常状态。博物馆和历史记录在建立恢复目标时具有不可替代的价值。保护成效评估应明确基线年代，并区分近期稳定与长期恢复。

DOI: <https://doi.org/10.1038/s41559-026-03074-6>

海洋与水生

27. Amplified Arctic iceberg traffic reshapes benthic biodiversity

(增强的北极冰山输运重塑底栖生物多样性)

Nature | 2026-06-10 | 【Recent】



图： Graphical abstract / article figure source: <https://doi.org/10.1038/s41586-026-10630-4>

Abstract

Seafloor observations and four decades of ship records link accelerating Arctic glacier disintegration to increased iceberg-delivered dropstones in the deep ocean. These hard substrates create patchy benthic habitats far from glacier fronts and reorganize biodiversity.

中文摘要

海底观测与四十年航行记录表明，北极冰川加速解体增加了冰山向深海输送的落石。这些硬质基底在远离冰川前缘处形成斑块化底栖栖息地，并重组生物多样性。

深度解读

该研究揭示冰冻圈变化向深海生态系统传播的一条具体路径。新增硬底会创造栖息地，但并不意味着气候变化产生净生态收益，因为群落结构和连通性同时被改变。深海监测应把冰山轨迹、落石密度和底栖演替结合起来。

DOI: <https://doi.org/10.1038/s41586-026-10630-4>

环境与气候

28. Mining triggers extensive additional deforestation in sub-Saharan Africa

(采矿在撒哈拉以南非洲触发大范围额外毁林)

Nature | 2026-06-03 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41586-026-10551-2>

Abstract

A difference-in-differences analysis of 16,627 mines estimates 187,000 ha of direct mining deforestation in sub-Saharan Africa during 2001–2020. For each hectare lost at mine sites, about 34 additional hectares were lost nearby through roads, settlements and agriculture.

中文摘要

研究对 16,627 处矿区开展准实验分析，估计 2001—2020 年采矿直接造成约 18.7 万公顷森林损失。每损失 1 公顷矿区森林，周边还会因道路、聚居和农业扩张额外损失约 34 公顷。

深度解读

关键发现是矿山边界外的间接影响远大于直接占地。能源转型矿产若不纳入供应链毁林核算，可能形成显著生态负担转移。环境影响评价应覆盖至少数十千米的空间外溢和建矿后的多年累积效应。

DOI: <https://doi.org/10.1038/s41586-026-10551-2>

海洋与水生

29. A 5.3-million-year-old deep-sea whale necropolis in the Diamantina Zone

(迪亚曼蒂纳带发现延续 530 万年的深海鲸落墓地)

Nature | 2026-06-10 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41586-026-10546-z>

Abstract

Submersible surveys documented five modern whale-fall communities and 476 fossil cetaceans across 1,200 km of the southeastern Indian Ocean floor. Isotopic dating indicates whale falls have persisted there for at least 5.3 million years.

中文摘要

载人潜水器在印度洋东南部约 1,200 千米海底范围记录 5 个现代鲸落群落和 476 具鲸类化石。同位素测年表明，该区域鲸落过程至少持续了 530 万年。

深度解读

鲸落既是短期化能合成生态系统，也是长期演化和生物地理档案。极低沉积速率与高密度鲸骨共同形成罕见的时间累积窗口。发现提示深海保护不能只关注热液和冷泉，也应纳入大型有机落物形成的生态热点。

DOI: <https://doi.org/10.1038/s41586-026-10546-z>

森林与生物多样性

30. Ecological multifunctionality of watersheds increases with tree species richness

(流域生态多功能性随树种丰富度增加)

Nature Communications | 2026-06-18 | 【Article in Press】



图：Graphical abstract / article figure source: <https://doi.org/10.1038/s41467-026-74914-z>

Abstract

A global watershed analysis reports that tree species richness is positively associated with multiple ecological functions. The result extends biodiversity–ecosystem–function relationships from plots to landscape-scale watersheds.

中文摘要

全球流域分析显示，树种丰富度与多项生态功能呈正相关，将生物多样性—生态系统功能关系从样地尺度扩展到景观和流域尺度。

深度解读

流域尺度证据更接近水源涵养、侵蚀控制和生产力管理的真实决策单元。相关性仍需结合因果设计判断，但结果支持在造林中避免单一树种配置。多功能目标下，树种组合与空间配置可能比单纯提高林地面积更重要。

DOI: <https://doi.org/10.1038/s41467-026-74914-z>

保护与生物多样性

31. Universal dose–response sensitivity of ecological processes to low levels of artificial light at night
(生态过程对低水平夜间人工光呈普遍剂量响应敏感性)

Biological Conservation | 2026-06-01 | 【Recent】



图： Graphical abstract / article figure source: <https://doi.org/10.1016/j.biocon.2026.111825>

Abstract

A synthesis of 15 datasets finds steep biological responses to artificial light below one lux across fish, birds, insects, plants and ecosystem processes. A common logarithmic dose–response model describes the cross-taxon pattern.

中文摘要

对 15 个数据集的综合分析发现，鱼类、鸟类、昆虫、植物及生态过程在低于 1 勒克斯的微弱人工光下已出现陡峭响应。跨类群变化可由共同的对数剂量—反应模型描述。

深度解读

最重要的管理含义是‘很暗’并不等于生态安全。传统照明标准多关注人眼和能耗，容易忽略亚勒克斯级暴露。自然保护区、河岸和迁徙廊道需要采用更严格的亮度、光谱和时段控制。

DOI: <https://doi.org/10.1016/j.biocon.2026.111825>

保护与生物多样性

32. Multi-species modelling to improve conservation decision-making for understudied taxonomic groups
(多物种模型改善研究不足类群的保护决策)

Biological Conservation | 2026-06-01 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1016/j.biocon.2026.111827>

Abstract

The paper shows how multispecies occupancy models borrow information across taxa to improve estimates for data-poor species. Compared with separate single-species models, they provide more stable and precise inputs for extinction-risk assessment.

中文摘要

研究说明多物种占域模型如何在物种间共享信息，从而改善数据贫乏物种的估计。相较逐物种建模，该方法能为灭绝风险评估提供更稳定、更精确的参数。

深度解读

保护名单常因数据不足而延迟，这类层级模型可把群落监测转化为可用证据。共享信息并不意味着物种相同，模型必须保留物种层差异并检查先验敏感性。对环境 DNA 和公民科学数据，该框架尤其有潜力。

DOI: <https://doi.org/10.1016/j.biocon.2026.111827>

保护与生物多样性

33. Beyond species richness: Habitat fragmentation reduces community occupancy and functional richness of mountain forest mammals

(超越物种丰富度：破碎化降低山地森林哺乳动物群落占域和功能丰富度)

Biological Conservation | 2026-06-01 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1016/j.biocon.2026.111828>

Abstract

Camera-trap analyses of 39 medium and large mammal species show that fragmentation reduces community occupancy and functional richness. Landscape configuration is more informative than forest cover alone, with nocturnal and specialist mammals especially vulnerable.

中文摘要

对 39 种中大型哺乳动物的相机监测分析显示，栖息地破碎化降低群落占域率和功能丰富度。景观配置比单纯森林覆盖率更关键，夜行性和专性物种尤其脆弱。

深度解读

面积相同的森林并不具有相同保护价值，斑块形状、隔离度和连接结构会筛选功能性状。只统计物种数可能掩盖功能同质化。山地保护应优先修复廊道和降低边缘效应，而非只追求覆盖面积。

DOI: <https://doi.org/10.1016/j.biocon.2026.111828>

岛屿与水库生态

34. Several small reservoir islands consistently support higher taxonomic, phylogenetic, functional, and interaction diversity than few large ones across multiple taxa

(多个小型水库岛屿在多类群和多维度上持续承载更高多样性)

Biological Conservation | 2026-06-01 | **【Recent】**



图：Graphical abstract / article figure source: <https://doi.org/10.1016/j.biocon.2026.111864>

Abstract

A multi-taxon reservoir-island study revisits the SLOSS debate using taxonomic, phylogenetic, functional and interaction diversity. Across these dimensions, several small islands consistently retain more diversity than a few large islands of equal total area.

中文摘要

一项多类群水库岛屿研究以分类、系统发育、功能和相互作用多样性重新检验 SLOSS 争论。在总面积相等时，多个小岛在这些维度上持续保留更多多样性。

深度解读

结果挑战把‘单一大斑块优先’机械应用于所有破碎景观。多个小岛可能通过环境异质性和物种互补提高总体多样性，但也更易受边缘效应影响。水库岛屿规划应在面积、连通性和跨岛互补之间做组合优化。

DOI: <https://doi.org/10.1016/j.biocon.2026.111864>

海洋与水生

35. Compounding impacts of riparian hardening and navigation erode phytoplankton community resilience in a large river ecosystem

(岸线硬化与航运的叠加影响削弱大型河流浮游植物群落韧性)

Journal of Environmental Management | 2026-06-15 | 【Recent】



图： Graphical abstract / article figure source: <https://doi.org/10.1016/j.jenvman.2026.130096>

Abstract

The study finds that riparian hardening and navigation jointly fragment phytoplankton association networks and reduce functional redundancy in a large river. Hardening primarily changes beta diversity, whereas navigation and seasonal gradients shape local alpha diversity.

中文摘要

研究发现岸线硬化与航运共同导致大型河流浮游植物关联网络破碎并降低功能冗余。岸线硬化主要改变 β 多样性，而航运和季节梯度更直接影响局地 α 多样性。

深度解读

河流管理若只控制营养盐，可能遗漏物理栖息地和扰动过程。网络破碎与功能冗余下降是系统失稳的早期信号。生态修复应把自然岸线、缓流区和航运强度纳入同一流域管理框架。

DOI: <https://doi.org/10.1016/j.jenvman.2026.130096>

恢复生态

36. Conservation and natural recovery limits of Madagascar's ultramafic rainforest

(马达加斯加超镁铁质雨林的保护价值与自然恢复上限)

Biological Conservation | 2026-06-01 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1016/j.biocon.2026.111867>

Abstract

Field baselines show exceptionally high woody-plant diversity in Madagascar's humid ultramafic rainforest, but seed rain and recruitment collapse after canopy removal. Shallow seed reserves dominated by pioneers make passive recovery after mining unlikely.

中文摘要

调查显示马达加斯加湿润超镁铁质雨林具有极高木本植物多样性，但冠层移除后种子雨和幼苗补充迅速崩溃。浅薄且以先锋种为主的土壤种子库意味着采矿后被动恢复很难成功。

深度解读

该研究清楚区分了‘停止扰动’与‘恢复生态系统’。特殊基质森林的物种库和土壤条件难以在矿后自然重建。管理上应优先避免损失，并把主动播种、表土保存和长期养护作为最低恢复条件。

DOI: <https://doi.org/10.1016/j.biocon.2026.111867>

海洋与水生

37. Linking trophic guild function to biodiversity and ecosystem resilience across river–floodplain–coastal continua (营养功能群连接河流—洪泛区—海岸连续体的多样性与韧性)

Estuarine, Coastal and Shelf Science | 2026-06-01 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1016/j.ecss.2026.109797>

Abstract

Diet, stable-isotope and biochemical evidence shows that floodplain cichlid trophic guilds use complementary energy pathways. Functional redundancy and cross-system trophic flows support nutrient cycling, organic-matter processing and downstream productivity.

中文摘要

食性、稳定同位素和生化证据表明，洪泛区慈鲷功能群利用互补的能量通道。功能冗余与跨系统营养流动共同支撑养分循环、有机质分解和下游生产力。

深度解读

该研究把鱼类多样性从物种清单转化为能量通道和生态功能。洪泛区与河口并非独立管理单元，上游水文改变会沿食物网传递到海岸渔业。维持季节性连通和功能群完整性是增强系统韧性的关键。

DOI: <https://doi.org/10.1016/j.ecss.2026.109797>
